

Exercise 2.7: Consider minimizing $f(x)$ for $x \in S$ where S is the set of integers. Prove that every point S is a local minimizer.

Proof. By definition for a point $x_* \in S$ to be a local minimizer of f , $f(x_*) \leq f(x)$ for all $x \in S$ such that $\|x - x_*\| < \epsilon$ for some $\epsilon > 0$. Let $x_* \in \mathbb{Z}$ and consider $\epsilon = .5$. By definition the only $x \in S$ such that $\|x - x_*\| < .5$ is x_* , therefore the statement $f(x_*) \leq f(x)$ is vacuously true and x_* is a local minimizer. $\square$

Exercise 3.1: Prove that the intersection of a finite number of convex sets is also a convex set.

Proof. Suppose S is a finite family of convex sets. Consider the set $A = \bigcap_{S_i \in S} S_i$. Let $0 \leq \alpha \leq 1$ and consider $a, b \in A$. Since a and b belong in the intersection of all sets S_i , they must also exist in every set S_i . Since S_i are convex it follows from the definition that $\alpha a - (1 - \alpha)b \in S_i$. Therefore $\alpha a - (1 - \alpha)b \in A$ and thus A is a convex set. $\square$

Exercise 3.3: Consider a feasible region S defined by a set of linear constraints,

$$S = \{x : Ax \leq b\}.$$

Prove that S is convex.

Proof. Consider $x, y \in A$. By definition $A(x) \leq b$ and $A(y) \leq b$. Consider some $0 \leq \alpha \leq 1$. Note that since α and $(1 - \alpha)$ are non-negative the following inequalities come out,

$$\alpha(Ax) \leq \alpha b,$$

$$(1 - \alpha)(Ay) \leq (1 - \alpha)b.$$

Adding both inequalities we get the following,

$$\alpha(Ax) + (1 - \alpha)(Ay) \leq (1 - \alpha)b + \alpha b,$$

$$A(\alpha(x) + (1 - \alpha)(y)) \leq (1 - \alpha + \alpha)b,$$

$$A(\alpha(x) + (1 - \alpha)(y)) \leq b.$$

Therefore by definition $(\alpha(x) + (1 - \alpha)(y)) \in S$. Thus S is convex. $\square$

Exercise 3.7: Let f be a convex function on a convex set $S \in \mathbb{R}^n$. Let k be a nonzero scalar, and define $g(x) = kf(x)$. Prove that if $k > 0$, then g is a convex function on S , and if $k < 0$, then g is a concave function on S .

Proof. Suppose f is a convex function on a convex set $S \in \mathbb{R}^n$ and let $g(x) = kf(x)$ for some $k > 0$. By definition of a convex function, we know that for all $0 \leq \alpha \leq 1$ and $x, y \in S$,

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$$

Since $k > 0$ multiplying both sides does not change the direction of the inequality,

$$\begin{aligned} kf(\alpha x + (1 - \alpha)y) &\leq k(\alpha f(x) + (1 - \alpha)f(y)), \\ g(\alpha x + (1 - \alpha)y) &\leq \alpha kf(x) + (1 - \alpha)kf(y), \\ g(\alpha x + (1 - \alpha)y) &\leq \alpha g(x) + (1 - \alpha)g(y). \end{aligned}$$

Thus $g(x)$ is also convex. Clearly when $k < 0$ the inequality switches sides and we get,

$$g(\alpha x + (1 - \alpha)y) \geq \alpha g(x) + (1 - \alpha)g(y).$$

Which by definition is concave. □

Exercise 3.13: Let f be a convex function on the convex set S . Prove that the level set $T = \{x \in S : f(x) \leq k\}$ is convex for all real number k .

Proof. Suppose f be a convex function on the convex set S , and consider the level set $T = \{x \in S : f(x) \leq k\}$ for some $k \in \mathbb{R}$. Since f is convex for all $0 \leq \alpha \leq 1$ and $x, y \in T$ we know that,

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$$

Since $x, y \in T$ we know that $f(x), f(y) \leq k$. By substitution,

$$\begin{aligned} f(\alpha x + (1 - \alpha)y) &\leq \alpha k + (1 - \alpha)k, \\ f(\alpha x + (1 - \alpha)y) &\leq \alpha k + k - \alpha k, \\ f(\alpha x + (1 - \alpha)y) &\leq k. \end{aligned}$$

Thus $\alpha x + (1 - \alpha)y \in T$ therefore by definition T is a convex set. □

Exercise 3.18: Express $(2, 2)^T$ as a convex combination of $(0, 0)^T$, $(1, 4)^T$, and $(3, 1)^T$.

Proof. Finding a convex combination of $(2, 2)^T$ can be done by solving the following equation,

$$\begin{pmatrix} 1 & 4 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 0 \\ 0 & 11 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 4 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ \frac{4}{11} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{6}{11} \\ \frac{4}{11} \end{pmatrix}$$

This gives us that z is free, $x = \frac{6}{11}$ and $y = \frac{4}{11}$. Since we know the magnitude of our solution needs to be 1 we can solve for the z within this constraint,

$$1 = \left(\frac{6}{11}\right) + \left(\frac{4}{11}\right) + z$$

$$\frac{1}{11} = z$$

□

Thus $(\frac{6}{11}, \frac{4}{11}, \frac{1}{11})$ is a valid convex combination.

Exercise 3.20: Determine if,

$$f(x_1, x_2) = 2x_1^2 + 3x_1x_2 + 5x_2^2 - 2x_1 + 6x_2.$$

is convex, concave, both or neither for $x \in \mathbb{R}^2$.

Proof. Recall the hessian definition of convex, given that at every point $x \in \mathbb{R}^2$

$$(x_1, x_2) \nabla^2 f(x_1, x_2)^T \geq 0.$$

Computing the hessian for the given $f(x)$,

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} 4 & -3 \\ -3 & 10 \end{bmatrix}$$

Computing it's eigenvalues,

$$\begin{vmatrix} 4 - \lambda & -3 \\ -3 & 10 - \lambda \end{vmatrix} = (4 - \lambda)(10 - \lambda) - 9 = \lambda^2 - 14\lambda + 31$$

Which gives us $\lambda_1 = 7 + 3\sqrt{2}$ and $\lambda_2 = 7 - 3\sqrt{2}$. Clearly the eigenvalues are larger than 0 so the hessian is positive definite, and therefore f is convex. $\square$

Exercise P5: For each of the following functions, determine if it is convex, concave, both or neither, on the real line $\mathbb{R}$. If the function is convex or concave, indicate whether or not it is also strict.

a. $f(x) = 8x - 15$

Solution:

This function is both concave and convex, since $f''(x) = 0$.

b. $f(x) = \sqrt{1 + x^2}$

Solution:

Note that,

$$f'(x) = \frac{x}{\sqrt{1 + x^2}}$$

$$f''(x) = \frac{1}{(1 + x^2)\sqrt{1 + x^2}}$$

This function on $\mathbb{R}$ is always positive and therefore the function is strictly convex.

c. $f(x) = 1/(2 + x^4)$

Solution:

Note that,

$$f'(x) = \frac{4x^3}{(2 + x^4)^2}.$$

$$f''(x) = -\frac{4x^2(-5x^4 + 6)}{(2 + x^4)^3}.$$

Solving for the roots of the second derivative in $\mathbb{R}$ we get $x = 0, (\frac{6}{5})^{1/4}, -(\frac{6}{5})^{1/4}$. To the left of $-(\frac{6}{5})^{1/4}$ the value of the second derivative is positive and to the right it's

negative. To the left and right $x = 0$ the second derivative is negative, and to the right of $x = (\frac{6}{5})^{1/4}$ the value of the second derivative is positive. Therefore the function is strictly convex on $(-\infty, -(\frac{6}{5})^{1/4}] \cup [(\frac{6}{5})^{1/4}, \infty)$ and concave from $[-(\frac{6}{5})^{1/4}, (\frac{6}{5})^{1/4}]$.

d. $f(x) = |x|$

This function is clearly convex.

e. $f(x) = 4 - 5x - 3x^2$

Computing the second derivative we get, $f''(x) = -6 < 0$. Therefore this function is strictly concave.

Exercise P6: Consider the scalar values function,

$$f(x_1, x_2) = \exp(-x_1^2 + 3x_1 - 2x_2 - x_2^2)$$

Compute the gradient and the Hessian of f . Find the location where f is maximum, and explain what properties of the gradient and Hessian show that it is a maximum.

Solution:

Computing the $\nabla f(x)$ we get the following,

$$\nabla f(x) = (f(x_1, x_2)(-2x_1 + 3), f(x_1, x_2)(-2x_2 - 2))^T.$$

Computing the Hessian $\nabla^2 f(x)$,

$$\begin{aligned} \nabla^2 f(x) &= \begin{bmatrix} -2f(x_1, x_2) + (-2x_1 + 3)^2 f(x_1, x_2) & f(x_1, x_2)^2(-2x_2 - 2)(-2x_1 + 3) \\ f(x_1, x_2)^2(-2x_2 - 2)(-2x_1 + 3) & -2f(x_1, x_2) + (-2x_2 - 2)^2 f(x_1, x_2) \end{bmatrix} \\ &= \begin{bmatrix} f(x_1, x_2)(4x_1^2 - 12x_1 + 7) & f(x_1, x_2)^2(4x_2x_1 - 6x_2 + 4x_1 - 6) \\ f(x_1, x_2)^2(4x_2x_1 - 6x_2 + 4x_1 - 6) & f(x_1, x_2)(4x_2^2 + 8x_2 + 2) \end{bmatrix} \end{aligned}$$

Finding where $\nabla f(x) = 0$, since any function e^x is nonzero, we must solve the following,

$$(-2x_1 + 3) = 0,$$

$$x_1 = \frac{3}{2}.$$

$$(-2x_2 - 2) = 0,$$

$$x_2 = -1.$$

Hence $\nabla f((\frac{3}{2}, -1)) = 0$. Plugging into the Hessian to determine if our value is a maximum we get,

$$\begin{aligned} \nabla^2 f((\frac{3}{2}, -1)) &= \begin{bmatrix} -2f(x_1, x_2) + 0 & 0 \\ 0 & -2f(x_1, x_2) + 0 \end{bmatrix} \\ &= \begin{bmatrix} -2f(x_1, x_2) & 0 \\ 0 & -2f(x_1, x_2) \end{bmatrix}. \end{aligned}$$

Clearly the values on the diagonals will be negative, and therefore the Hessian at point $(\frac{3}{2}, -1)$ is negative definite. Thus the point $(\frac{3}{2}, -1)$ is a maximum.

Exercise P7: a. Consider the vector-values function,

$$f(x_1, x_2, x_3) = \begin{pmatrix} x_1^2 + x_2^2 + x_3^2 - 1 \\ x_2 - \arctan(x_1) \\ x_3^3 - x_3 - x_2 \end{pmatrix}$$

Compute the Jacobian of f .

Solution:

Computing the Jacobian $\nabla f(x)^T$,

$$\nabla f(x)^T = \begin{pmatrix} 2x_1 & 2x_2 & 2x_3 \\ -\frac{1}{1+x_1^2} & 1 & 0 \\ 0 & -1 & 3x_3^2 - 1 \end{pmatrix}$$

- b. The Jacobian of f in part (a) is a 3×3 matrix, just like the Hessian of a scalar function. Is the Jacobian of f in part (a) actually the Hessian of some scalar function? This question can be answered by considering the symmetry of a matrix for example point $x = (1, 1, 1)$ for concreteness.

Solution:

Since the Hessian defined as the following,

$$[\lambda^2 f(x)]_{ij} = \frac{d^2 f(x)}{dx_i dx_j}$$

it must always be a symmetric matrix because of the property that mixed partial derivatives are equal,

$$\frac{d^2 f(x)}{dx_i dx_j} = \frac{d^2 f(x)}{dx_j dx_i}.$$

Clearly the Jacobian computed in (a) is not symmetric for all $x \in \mathbb{R}^n$ since $[\nabla f(x)^T]_{3,2} = -1$ and $[\nabla f(x)^T]_{2,3} = 0$. Thus it cannot be a Hessian for some scalar function.